

Mutation free-energy labels improve low-data nanobody thermal stability prediction

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Thermal stability is a key constraint in nanobody development, but experimental melting-temperature (T_m) measurements are often sparse. Computational methods can label many more sequence variants, although they usually report quantities related to T_m rather than T_m itself. We tested whether such labels can improve low-data nanobody T_m prediction when model selection is based only on experimental T_m validation. Using a public benchmark with 57 training, 114 validation, and 396 held-out test sequences, we trained multi-task transfer models in which the T_m predictor shared a protein-language-model sequence representation with one computational source task at a time. The source labels included mutation free energies, structure-based mutation scores, Rosetta scores for protein-language-model proposal sets, and molecular-dynamics trajectory summaries. Alchemical free-energy perturbation (FEP) gave the clearest benefit: held-out T_m mean absolute error decreased from 6.61 to 6.26°C compared with training only on experimental T_m labels, a small but consistent paired improvement of 0.35°C (90% confidence interval, 0.24–0.46°C). The same gain was not reproduced by increasing the ESM2 encoder size or by using an MD-derived Q-value. Rosetta-scored proposal sets could be used as source labels, but their advantage over random variants remained exploratory. These results support the use of computational labels for nanobody T_m prediction when their physical meaning transfers to the experimental endpoint.

nanobody thermostability | melting temperature | simulation-to-real transfer learning | protein language model | free energy perturbation
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Introduction

Simulation data for low-data experimental prediction

Machine learning has made it possible to screen large molecular and materials spaces before costly experimental testing, but its usefulness still depends on the availability of reliable property labels. In materials informatics, this data bottleneck has motivated a simulation-to-real (Sim2Real) strategy: large computational databases generated by first-principles calculations, quantum chemistry, or molecular dynamics simulations are used as source data, and the resulting models are transferred to smaller experimental target datasets (1–3). A recent study in polymers and inorganic materials showed that the downstream prediction error of transferred models can scale with the amount of com-

putational data, providing a quantitative way to judge whether a growing simulation database has real experimental value (3). This framing is important because simulation data are not useful by default: simulations and experiments are separated by model assumptions, sampling limits, and systematic domain gaps.

Protein engineering faces an analogous problem, but with an additional layer of physical mismatch. Experimental measurements of protein properties are expensive and sparse, whereas biomolecular simulations and physics-based calculations can generate labels for many designed or mutated sequences. The challenge is sharper than in many property-matched transfer settings, because the simulated quantity may not be the same as the experimental target. For example, a target phenotype may be the melting temperature (T_m), the temperature at which a protein loses half of its folded population under a given assay condition. A tractable calculation, by contrast, may estimate how much a mutation changes stability, assign a structure-based stability score, or summarize a molecular-dynamics trajectory. The central question is therefore not simply whether simulation data can be added to a training set, but which simulated physical quantities can transfer to the experimental phenotype.

Nanobody thermostability in the high-throughput era

We examine this question using nanobody thermostability as a concrete case. Nanobodies, or single-domain VHH antibodies, are attractive engineering scaffolds because of their small size, modularity, and binding properties (4, 5). Their practical use, however, depends on thermal stability during expression, storage, formulation, and downstream screening. Experimental T_m measurements remain limited, and direct molecular simulation of thermal stability is computationally demanding (6). Sequence-based predictors and protein language models (PLMs) offer an attractive alternative: PLMs such as ESM2 learn broadly useful protein representations from large sequence corpora (7, 8), and recent methods have used deep sequence representations to predict protein or nanobody melting temperatures (9–11). High-throughput stability measurements are also changing the data landscape. Tsuboyama et al. used cDNA-display proteolysis to generate approximately 776,000 folding-stability measurements for natural and designed

protein domains (12). This advance shows that experimental ΔG can be measured at unprecedented scale for suitable small domains, but it does not yet solve the data problem for full-length nanobody VHH domains: the curated mega-scale domains span 40–72 amino acids, whereas nanobody VHH domains are typically much longer, and comparable high-throughput ΔG measurements are not yet available for such scaffolds. Moreover, a model fine-tuned on the mega-scale dataset transferred less well to 177–501-residue proteins than to small domains (13). Large experimental stability resources are therefore powerful, but target-specific supervision and careful validation remain essential for nanobody Tm prediction (14–16).

Study design and main finding

Here we ask whether simulation-derived labels can provide that additional supervision when experimental Tm labels are scarce. We treat Tm prediction as the experimental target task and treat each computational label as a source task that can shape the learned sequence representation. The implementation uses ESM2, a pretrained protein language model, as the shared sequence encoder (8). The computational labels represent several kinds of computational information: mutation free-energy changes from alchemical free-energy perturbation (FEP) calculations, mutation scores from Rosetta, stability scores from ThermoMPNN, Rosetta scores on variants proposed by ESM2 or sampled at random, and a molecular-dynamics (MD)-derived Q-value measuring native-contact persistence. To keep the comparison target-centered, all model settings are selected using an experimental validation set, and all final claims are evaluated on the same held-out experimental Tm test set with paired error comparisons. This design prevents high performance on a computational task from being mistaken for improved Tm prediction.

The main result is that the benefit of simulation-derived supervision is source-dependent. Mutation free-energy labels from FEP provide the clearest improvement over training on Tm labels alone, whereas the MD-derived Q-value does not improve held-out Tm prediction under the same evaluation design. Rosetta-scored ESM2 proposal sets provide a weaker exploratory signal, suggesting a possible connection to future generator–physics–predictor design loops without demonstrating closed-loop sequence optimization. Taken together, these results extend the Sim2Real argument from materials informatics to a biomolecular setting in which the source and target labels differ in physical meaning. Useful simulation data are not merely abundant; they encode a physical quantity that transfers to the experimental property of interest.

Materials and Methods

Experimental and technical design

This study was designed to test whether computational source labels can improve experimental nanobody melting-temperature (Tm) prediction when the source label is physically related to, but not identical with, Tm. We therefore used a multi-task transfer-learning design: experimental Tm prediction was the target task, and computational labels were source tasks that shared part of the neural network with the target task. The central constraint was that all model selection and all final claims were anchored to experimental Tm performance. Source-task losses were used for representation learning during training, but source-task validation errors were not used to select final checkpoints or final hyperparameters.

Experimental Tm data set and split definitions

Experimental Tm labels were taken from a public nanobody benchmark data set, NbBench, which provides a thermo-seq task with predefined splits (15). The processed files used in this study contained 57 training sequences, 114 validation sequences, and 396 test sequences.

Split definitions. The training split was used to fit model parameters. The validation split was used for checkpoint selection, early stopping, and hyperparameter selection. The test split was held out until the selected settings were fixed and was used only for final reporting. This definition applies to both Tm-only models and multi-task transfer models.

Target-label scaling. The target CSV files contained an amino-acid sequence column and a Tm label in degrees Celsius. During training, Tm labels were transformed to a scaled regression target. The scaler was fit only on the 57 training labels, using a robust scaling step followed by min–max scaling to [0,1]. The fitted scaler was then applied to validation labels during training. At evaluation time, model predictions were inverse-transformed back to degrees Celsius before computing errors. The validation and test label distributions were therefore not used to fit the target-label scaling transform.

Experimental-label scaling reference. For the experimental-label scaling reference, the Tm training split was subsampled to the requested number of labels using the run seed. The validation and test splits were not subsampled. These experiments provide a target-label reference for interpreting whether computational source-label scaling behaves in the expected direction.

Computational source-label data sets

Computational labels were represented as source tasks rather than as additional Tm measurements.

Source-label classes. The source-label collection comprised mutation free-energy changes from alchemical free-energy perturbation (FEP), structure-based

mutation-effect scores from Rosetta, learned mutation-effect scores from ThermoMPNN, Rosetta scores assigned to ESM2-proposed and random variants, and a molecular-dynamics (MD)-derived Q-value label that summarizes native-contact persistence. All source labels were preprocessed to [0,1]-scaled regression targets and were used only through their own source-task heads.

Processed source-table sizes. Mutation-effect sources were organized as two template-specific tables derived from the 1MEL and 4IDL nanobody structures. FEP, Rosetta, and ThermoMPNN each contained 435 and 409 processed rows for the two templates. Variants proposed by ESM2 and random variants scored by Rosetta each contained 1000 and 1000 processed rows. The MD-derived Q-value source was generated from a separate panel of simulated single-domain antibody structures selected from SAbDab, the Structural Antibody Database (17). The processed MD table contained 1143 analysis-ready rows. Row counts exclude headers.

Sequence-overlap check. We checked exact sequence overlap between source-label tables and the NbBench splits. The mutation-effect source tables and Rosetta-scored variant source tables with explicit sequence columns had no exact overlap with the NbBench training, validation, or test sequences. The SAbDab-derived MD table contained a small number of exact matches to NbBench sequences, including eight test-set sequences; because the MD Q-value source did not improve final Tm prediction, this overlap does not support the positive FEP result but is reported for transparency.

Source-label sampling. For a mutation-effect experiment with requested source-label count n , up to n rows were sampled independently from each of the two template-specific source tables using the run seed. The two template-specific tables were assigned separate task identifiers. For an MD-derived Q-value experiment with requested source-label count n , n rows were sampled from the single processed Q-value table. After sampling, source-label rows were split 80/20 into source-training and source-validation rows using the same run seed. These source-validation rows were retained for monitoring and control analyses. In the reported final protocol, however, the validation data passed to the trainer for checkpoint selection and early stopping were filtered to the experimental Tm validation rows.

SAbDab-derived MD simulations and Q-value labels

The MD source-label set was built from nanobody structures selected from a SAbDab-derived summary table (17). We selected one heavy-chain nanobody structure per PDB entry when an IMGT-renumbered coordinate file was available and the selected chain could be verified in that file. The 400 K simulation panel contained 1146 nanobody systems. After requiring a usable trajectory, topology, reference structure, sequence of at least 50 residues, and at least one selected native contact, 1143 trajectories entered the processed source-label ta-

ble.

MD systems were prepared from the selected nanobody chain only. The IMGT PDB file was used as the structure source, the heavy chain was extracted, non-protein components were excluded, termini were left uncapped, and protein protonation was assigned at pH 7.4. The preparation workflow used PDBFixer for structure cleaning and missing-atom handling, followed by pdb2pqr/PROPKA for pH-aware protonation and Amber-compatible atom naming when available (18, 19). Systems were solvated in explicit OPC water with 15 Å padding and 0.15 M NaCl, and Amber topologies were built with the ff19SB protein force field and OPC water model (20–22).

The trajectory protocol used OpenMM on CUDA devices (23). Each system was first equilibrated at 300 K with 1 ns NVT followed by 1 ns NPT at 1 atm, using a 2-fs timestep without hydrogen-mass repartitioning. The 400 K branch was then generated from the 300 K equilibrated state by two restrained NVT relaxation stages: 1 ns at 400 K with C α restraints followed by a second 1 ns 400 K NVT restrained relaxation. The production trajectory used 100 ns of restraint-free 400 K NVT dynamics, hydrogen-mass repartitioning with 4 amu hydrogen mass, a 4-fs timestep, Langevin middle integration with 1 ps⁻¹ collision rate, PME electrostatics with a 1.0-nm nonbonded cutoff, and coordinate/energy reporting every 100 ps. The reported MD source labels use the 400 K production branch.

Q-values were extracted from the 400 K production trajectories with MDTraj (24). For each trajectory, the simulation topology was loaded from the Amber parameter file and the native reference was defined as frame 0 of the production trajectory. A smooth native-contact fraction (25) was computed from backbone-heavy-atom contacts and averaged over the terminal portion of the trajectory. The raw trajectory averages were min-max scaled to [0,1] before training. The exact contact selection, contact function, averaging window, and processed-column convention are reported in Supplementary Methods.

FEP mutation free-energy labels

The FEP source labels were mutation free-energy changes for variants associated with the 1MEL and 4IDL nanobody template sequences. FEP calculations followed the same protocol as our related nanobody thermostability study. Briefly, alchemical mutation simulations were run with NAMD 3 using GPU acceleration (26). Input systems were prepared in VMD (27) using a customized AlaScan workflow (28). Simulation coordinates were initialized from AlphaFold 3 models of the corresponding wild-type sequences (29). Systems used the CHARMM36 protein force field (30), TIP3P water (31), and 150 mM NaCl.

Each alchemical transformation used a dual-topology mutation setup with a hybrid-residue topology for CHARMM36. The transformation was divided into 20

equally spaced λ windows from wild type to mutant. Forward and backward transformations were simulated, with 1 ns per direction, a 2-fs integration timestep, and 200 ps equilibration per window before data collection. Simulations were performed in the NPT ensemble at 300 K and 1 atm with Langevin temperature control and a Langevin piston barostat. Bonds involving hydrogen atoms were constrained with SHAKE (32), and long-range electrostatics were evaluated with particle mesh Ewald (33). Relative free energies were estimated with the Bennett acceptance ratio (34). The processed learning labels used the sign convention and [0,1] scaling stored in the FEP CSV files. Specifically, the processed tables retain the raw mutation free-energy value, its negated value, and the min-max scaled source label used by the neural network. Under this convention, larger processed FEP labels correspond to more favorable mutation free energies after negation and scaling.

Rosetta and ThermoMPNN mutation-effect labels

Rosetta mutation-effect labels were generated with the Rosetta ddg_monomer application (35). For each template, the PDB structure was cleaned to the target chain, variant sequences were compared with the corresponding wild-type sequence, and the sequence differences were converted to Rosetta mutation-list files. The calculation used full-atom Rosetta with the ref2015 score function (36), 50 iterations, Cartesian minimization enabled, local-only optimization disabled, side-chain-minimization-only disabled, minimum-score aggregation, and no PDB dumping. The same Rosetta protocol was used for the template mutation-effect tables, the variants proposed by ESM2, and the random-variant control.

For the ESM2-proposed variant source, ESM2 650M was used to sample 100,000 two-mutation variants from each wild-type template sequence, variants were ranked by pseudo-log likelihood, and the top 1% were retained for Rosetta scoring (8). The random-variant control used approximately 1000 two-mutation variants constructed by randomly choosing mutation positions and amino acid substitutions, followed by the same Rosetta scoring workflow.

ThermoMPNN source labels were computed from the same template-specific variant sets using ThermoMPNN, a stability-change predictor trained by transfer learning from larger protein-stability data (37). The ThermoMPNN outputs were converted to the same processed CSV format as the other mutation-effect source tasks.

Neural-network architecture

All models used an ESM2 protein language-model encoder (8) followed by lightweight regression heads. Unless otherwise stated, the encoder was the 8M-parameter ESM2 model. Sequences were tokenized with truncation and padding to a maximum length of 160 residues. The

first-token encoder representation was passed to a shared multilayer perceptron with hidden dimensions 256, 128, and 32, ReLU activations, and dropout. A scalar Tm head predicted the scaled target Tm label. In source-transfer models, additional scalar source heads predicted the source-task labels from the same shared representation.

The Tm-only baseline used the ESM2 encoder, the shared trunk, and the Tm head with only experimental Tm labels. Source-transfer models used the same target path plus source-task heads. The main FEP model used separate source heads for the two template-specific mutation-effect tasks, which was the best-performing source-head design in the source-head comparison. Additional controls evaluated a shared mutation-effect head, a shared head conditioned on a template embedding, and a shared head with template-specific affine calibration. Architecture controls also evaluated residual and latent source-fusion variants, but the main result used the original shared-trunk architecture.

Hot-encoder and frozen-encoder training

Two encoder-training regimes were evaluated. In the hot-encoder regime, the ESM2 encoder and regression heads were optimized jointly. The encoder used a separate learning rate from the newly initialized trunk and heads, so that the pretrained representation was updated more conservatively than the output layers. In the final FEP setting, the selected encoder learning rate was 3×10^{-5} and the head learning rate was the default 3×10^{-4} .

In the frozen-encoder control, all ESM2 parameters were fixed by setting their gradients off. The encoder was run once to precompute sequence embeddings for the target and source examples, and training then optimized only the downstream trunk and task heads. This control asks whether the source labels help when the pretrained sequence representation is used purely as a fixed feature extractor. Encoder-size controls repeated selected Tm-only and FEP-assisted settings with 35M- and 650M-parameter ESM2 encoders.

Loss function and multi-task objective

Each training example was represented as (x_i, y_i, t_i) , where x_i is the sequence, y_i is the scaled regression label, and t_i is the task identifier. The target Tm task used one task identifier, and each source task used its own identifier. A minibatch could contain target and source examples, but each example contributed loss only through the head for its task.

The per-task regression loss was Huber loss with $\delta = 1.0$ on scaled labels. The default multi-task objective used learnable task-uncertainty weights (38). For task k with loss L_k and learned log-scale s_k , the contribution was

$$\frac{L_k}{2 \exp(2s_k)} + s_k.$$

The total loss was the sum over the tasks present in the minibatch. Fixed source-task weights were also evaluated as controls, particularly for the MD-derived source labels.

Training, early stopping, and hyperparameter selection
Models were trained with AdamW, cosine learning-rate scheduling, batch size 16, weight decay 0.04, 100 warmup steps, and at most 400 epochs. The default head learning rate was 3×10^{-4} , the default hot-encoder learning rate was 1×10^{-4} , and the default dropout rate was 0.15. Training used mixed precision on CUDA devices. Checkpoints were saved and evaluated once per epoch.

Within each run, the best checkpoint was the checkpoint with the lowest mean squared error on the experimental Tm validation examples. Early stopping used the same validation metric with a patience of 15 epochs. In source-transfer models, source-label rows were present in the training set, but they were removed from the validation data passed to the trainer for checkpoint selection and early stopping. This is the implementation-level step that ensures that the reported checkpoints are selected by target-task validation performance rather than by source-task performance.

Hyperparameter selection was also performed on the experimental Tm validation split. Candidate settings varied encoder learning rate, head learning rate, dropout, loss-weighting mode, source-head design, and selected architecture controls. The search included encoder learning rates of 3×10^{-5} , 1×10^{-4} , and 3×10^{-4} ; a lower head learning rate of 1×10^{-4} paired with encoder learning rate 3×10^{-5} ; dropout rates of 0.05, 0.15, and 0.30; and fixed source-task loss weights where relevant. Each candidate setting was evaluated as a three-seed ensemble on the Tm validation split. The selected setting for each source class was then evaluated on the held-out test split as a ten-seed ensemble.

Independent run seeds controlled model initialization, target-training subsampling in the target-label scaling experiment, source-label sampling, and source-task train/validation splitting. Final comparisons used seeds 1–10. The label-count curves used fixed source-specific training recipes while varying the number of source labels, rather than retuning a separate model for every label-count point.

Final evaluation and statistics

Final evaluation used the same 396 held-out Tm test examples for all conditions. For each condition, the ten independently seeded models produced scaled Tm predictions for every test sequence. Predictions were averaged in scaled-label space and then inverse-transformed to degrees Celsius. The primary metric was mean absolute error (MAE) in degrees Celsius. Root mean squared error, coefficient of determination, Spearman correlation, and Pearson correlation were also computed, but MAE was used for the main comparisons because it is directly

interpretable as average temperature error.

Confidence intervals were estimated by nonparametric bootstrap resampling over held-out test proteins. For a single condition, the absolute-error vector from the ten-seed ensemble was resampled with replacement 10,000 times using a fixed random seed. The reported 90% confidence interval is the 5th to 95th percentile range of the bootstrapped MAE values.

For paired comparisons, the same bootstrap indices were applied to the absolute-error vectors from the candidate and reference conditions. The candidate-minus-reference MAE difference was then computed for each bootstrap sample. This paired bootstrap preserves the matched-test-example structure of the comparison. In figures and result tables, ΔMAE is candidate minus reference, so negative values indicate lower error than the reference. The one-sided tail probability computed by the evaluation code is the fraction of paired bootstrap samples in which candidate-minus-reference ΔMAE is greater than zero. The manuscript emphasizes effect sizes and 90% confidence intervals.

Implementation details

The training and evaluation pipeline constructs target and source tasks, applies target-label scaling, performs seeded source-label sampling, filters validation data for target-task checkpoint selection, evaluates seed ensembles, and writes bootstrap statistics. Each run writes a machine-readable summary containing the command-line arguments, resolved model settings, per-point metrics, bootstrap intervals, and paired comparisons. These summaries are the direct inputs to the figure-generation workflow and will be archived with the processed data for reproducibility.

Results

A target-centered framework for using simulation labels in Tm prediction

Framework. We first established an evaluation framework for asking whether simulation-derived labels improve experimental Tm prediction. The target task was nanobody Tm regression from amino-acid sequence. Computational labels were treated as source prediction tasks rather than as substitutes for experimental Tm. Each model used a pretrained protein sequence model, ESM2, to convert a sequence into a learned representation. The representation was shared by a Tm prediction head and a computed-label head, allowing the computational task to influence the representation while keeping the final objective anchored to experimental Tm prediction (Fig. 1).

Target-centered evaluation. All candidate settings were selected using an experimental Tm validation set, and the final numbers were computed only on a held-out experimental Tm test set. Improvements were assessed by paired differences in absolute error on the same test examples. This design addresses a common ambiguity

in simulation-to-experiment learning: a computational task can be learnable and physically meaningful without improving the target experimental phenotype.

Scaling depends on the physical label

Label-count sweeps. We next varied the number of experimental and computational labels to determine whether more source data translated into better Tm prediction. As expected, increasing the number of experimental Tm labels improved the model trained only on Tm measurements (Fig. 2). The target-label sweep used 10, 20, 30, 40, and 57 experimental Tm training labels. The FEP sweep used 10, 40, 80, 160, and 320 rows per template-specific source table, and the MD Q-value sweep used 10, 40, 80, 160, 320, and 640 rows from the single MD source table. Mutation free-energy labels from FEP calculations showed a beneficial trend: the largest FEP label-count setting tested gave the best held-out Tm error among the FEP settings. The curve was not strictly monotonic, so we do not interpret it as a clean power law. Instead, the largest tested setting is consistent with a benefit from additional mutation free-energy labels when models are selected and evaluated through the experimental Tm endpoint.

MD-derived Q-value scaling. The MD-derived Q-value label behaved differently. This label measured how strongly native contacts persisted during high-temperature MD trajectories. Increasing the number of these labels did not yield a robust improvement over learning from Tm measurements alone, even though the computed labels were also generated from simulations. The label-count experiment therefore points to the central comparison tested below: the number of computed labels is not sufficient by itself; the transferable physical content of the label matters.

All-atom simulation labels differ in their transfer to Tm prediction

All-atom source-label comparison. We then focused on two source labels derived from all-atom simulations. Training on the experimental Tm labels alone achieved a held-out test MAE of 6.61°C. Adding FEP mutation free-energy labels reduced the MAE to 6.26°C, corresponding to a small but consistent paired Δ MAE of -0.35°C (90% confidence interval, -0.46 to -0.24°C; Fig. 3). Negative paired changes indicate lower error than the Tm-only reference. In contrast, the MD-derived Q-value worsened the final test error to 6.73°C (paired change, +0.12°C; 90% confidence interval, +0.01 to +0.22°C). Both labels came from atomistic simulations, but only the mutation free-energy label improved the experimental endpoint.

Encoder adaptation. The encoder controls supported this source-label distinction. In a frozen-encoder control, overall errors increased, but FEP still improved over the frozen experimental-only baseline: frozen Tm-only learning achieved 7.38°C MAE, whereas frozen FEP

achieved 7.20°C MAE. The improvement was smaller than in the hot-encoder setting, and the frozen FEP model remained worse than the hot-encoder FEP model. The MD Q-value did not provide the same benefit in either encoder regime. These results indicate that FEP labels carry useful information even when the representation is fixed, but encoder adaptation is important for extracting the largest target-task benefit observed here.

The FEP gain is not explained by sequence-model size alone

Encoder-size control. Because the size of a protein language model can affect downstream performance, we tested whether the benefit of FEP labels could be reproduced simply by using a larger sequence encoder. It could not. In the experimental-only setting, larger 35M and 650M encoders did not improve held-out performance relative to the 8M encoder. In the FEP setting, the 35M and 650M encoders reached 6.50 and 6.65°C MAE, respectively, both worse than the 8M FEP model. FEP labels remained beneficial within each size comparison, but the best absolute performance was obtained with the smallest encoder and FEP labels. Within the tested settings, the gain was therefore associated with the physical source label rather than with scaling the language model (Supplementary Fig. 4).

Structure-based labels provide weaker source signals

Other structure-based source labels. We next compared computational labels that did not come from the FEP or MD Q-value pipelines. Rosetta scores for ESM2-proposed variants reached 6.45°C MAE (paired change, -0.17°C; 90% confidence interval, -0.31 to -0.03°C), and ThermoMPNN stability scores reached 6.46°C MAE (paired change, -0.15°C; 90% confidence interval, -0.30 to -0.01°C; Fig. 4). Rosetta mutation scores reached 6.53°C MAE (paired change, -0.09°C; 90% confidence interval, -0.20 to +0.02°C), and random variants scored by Rosetta reached 6.51°C MAE (paired change, -0.10°C; 90% confidence interval, -0.22 to +0.03°C). These sources were weaker than FEP, and the confidence intervals for the two Rosetta controls crossed zero.

Proposal-set comparison. The ESM2-proposed variants scored by Rosetta were slightly better than random variants scored by Rosetta, with a paired MAE difference of -0.07°C in favor of the ESM2-proposed set. The confidence interval crossed zero, so this comparison should be interpreted as exploratory rather than as evidence that the present generator improves variant selection. It nevertheless shows that sequence proposals can be annotated by physics-based scoring and used as source labels for the Tm predictor, a feasibility point revisited in the Discussion.

Overall pattern. Across the source labels tested here, the strongest and most consistent improvement within this benchmark came from FEP mutation free energies.

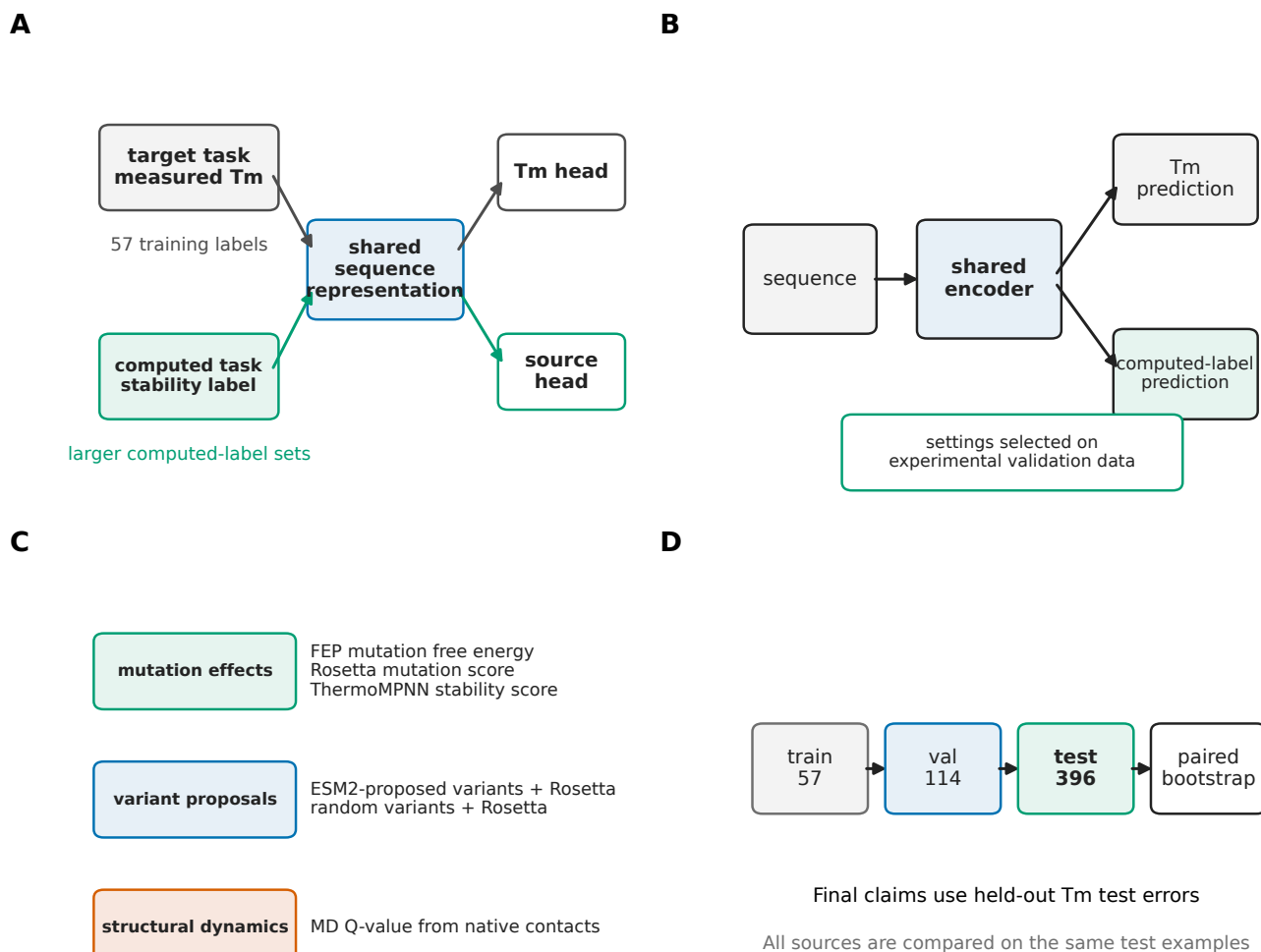


Fig. 1. Multi-task transfer-learning framework for using simulation data in experimental Tm prediction. (a) Experimental Tm labels are scarce, whereas simulation and physics-based calculations can provide larger computed-label pools. (b) Nanobody Tm prediction is used as a concrete case in which the computed labels are related to, but not identical with, the experimental target. (c) The model shares a pretrained protein sequence representation between the Tm target head and a computed-label head. (d) Model settings are selected on an experimental validation set, and final claims are evaluated on held-out experimental Tm test examples using paired error statistics.

Other structure-based labels and proposal-set scores produced weaker gains, while the MD-derived Q-value did not improve held-out Tm prediction. The resulting pattern is selective rather than additive: computational labels help only when the information they provide transfers to the experimental Tm endpoint.

Discussion

Source-label transfer is selective

This study asked how simulation-derived labels should be used when the experimental target is scarce and the simulated quantity is related to, but not identical with, that target. Using nanobody Tm prediction as a case study, we found that FEP mutation free-energy labels produced a small but consistent improvement in held-out experimental Tm prediction. The same improvement was not reproduced by the MD-derived Q-value

descriptor tested here, nor was it explained by using a larger ESM2 encoder. These findings support a selective view of Sim2Real learning for biomolecules: simulation data are useful when the source label encodes physical information that transfers to the experimental phenotype under target-centered validation.

Mutation free-energy labels connect relative and absolute stability

The helpful source label did not measure Tm itself. FEP estimates mutation free-energy changes, whereas Tm is an experimental phenotype reflecting the temperature at which folded and unfolded states are balanced under a particular assay condition. These are not interchangeable quantities. Nevertheless, both describe the same underlying stability landscape from different perspectives. Sparse Tm measurements anchor absolute thermal-stability values for observed sequences, while

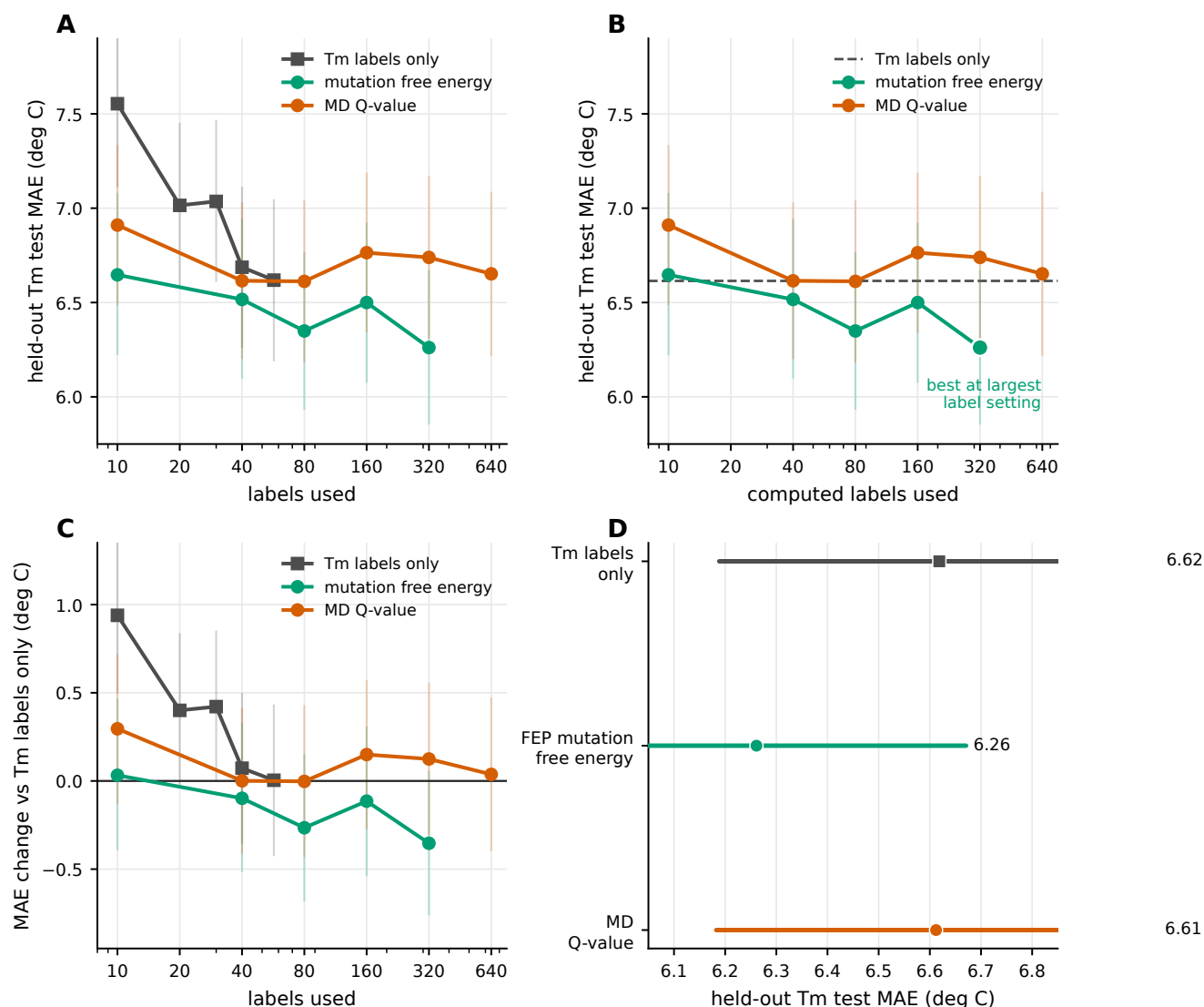


Fig. 2. Label-count scaling of experimental and computational supervision. (a) Experimental Tm-label scaling, FEP mutation free-energy scaling, and MD Q-value scaling are plotted together to compare the direction of label-count effects. (b) Computed-label scaling for FEP and the MD Q-value, with the Tm-label-only reference shown as a dashed line. (c) The same curves shown as MAE changes relative to the Tm-label-only reference. In panels a–c, points show ensemble MAE values and thin vertical intervals show 90% bootstrap intervals. (d) Best points from the label-count sweeps are replotted as interval estimates to compare the experimental-only, FEP, and MD Q-value references.

mutation free-energy labels provide local directions in sequence space. The FEP improvement therefore suggests that source labels can help even when they are not direct surrogates for the target, provided that they describe a physically relevant perturbation of the target property.

Scaling behavior depends on the source label

These results extend the materials-informatics Sim2Real argument to a more heterogeneous biomolecular setting. In materials applications, transfer learning often links computational and experimental estimates of related material properties, and scaling behavior can quantify the downstream value of a growing computational database (3). In the present protein case, the most useful computational labels were not additional Tm measurements but mutation free energies. The label-count sweep

for FEP was consistent with a benefit at the largest tested setting, but it was not smooth enough to claim a universal biomolecular scaling law. The stronger conclusion is source dependence: simulation-derived data should be evaluated by target-task performance, and source labels must be chosen for transferable physical content rather than availability alone.

Model-size controls

The encoder controls sharpen this conclusion. Prior work on PLMs has shown that fine-tuning can be critical for downstream protein-property prediction (16), and related nanobody Tm analyses suggest that larger pretrained ESM2 models do not automatically produce better Tm predictors. Our own size controls reached the same practical conclusion: 35M and 650M ESM2 encoders did not outperform the 8M encoder in either

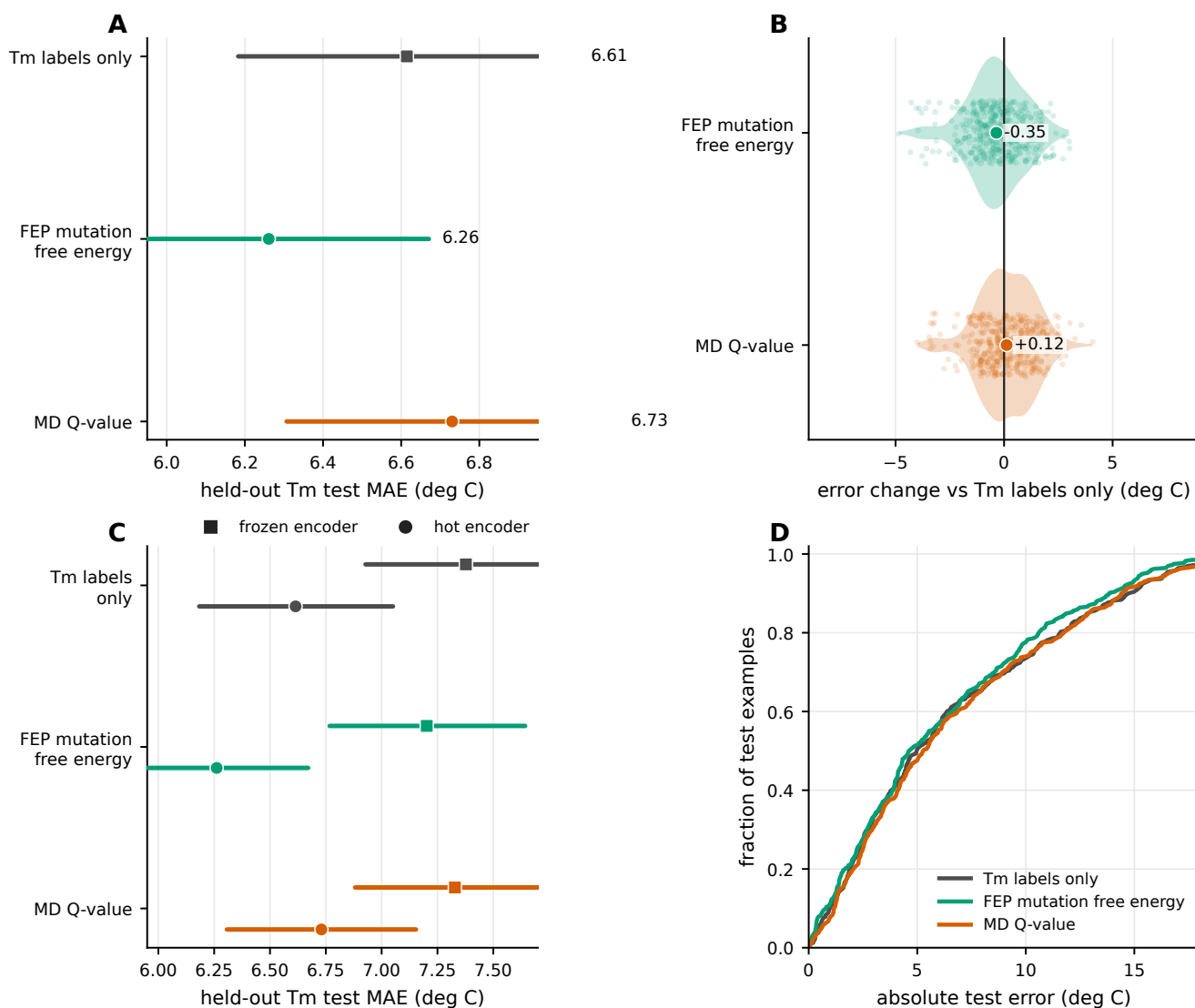


Fig. 3. All-atom simulation labels differ in their transfer to Tm prediction. (a) Final test MAE for experimental Tm labels alone, FEP mutation free-energy labels, and the MD-derived Q-value. Intervals show bootstrap confidence intervals over test-set absolute errors. (b) Per-example and paired changes in absolute error relative to training on Tm labels alone. Negative values indicate lower error than the reference. (c) Frozen-encoder and hot-encoder comparisons for the same three conditions. (d) Cumulative distributions of held-out absolute errors.

Tm-only or FEP-assisted settings. The best model used the 8M encoder with FEP source labels. Thus, in this dataset, increasing language-model capacity did not substitute for source labels carrying information about sequence-dependent stability changes.

Encoder adaptation and descriptor choice

The frozen-encoder experiment further indicates how the source-label signal is used. FEP labels improved performance even when the encoder was fixed, showing that part of the signal can be captured by the existing sequence representation. However, the best FEP model required updating the encoder. This suggests that FEP labels help most when they are allowed to reshape the representation toward stability-relevant sequence directions while still being selected by experimental Tm performance. In practical terms, the result argues for

controlled encoder adaptation rather than either fully fixed feature extraction or unconstrained source-task optimization.

The MD-derived Q-value result should be interpreted with the same specificity. It should not be read as evidence that MD-derived descriptors are generally unhelpful. Rather, this particular high-temperature native-contact summary did not transfer to Tm prediction under the present model-selection protocol. The descriptor may still capture aspects of structural durability, but it compresses each trajectory to a single native-contact persistence statistic and may miss mutation-specific energetic effects that determine how sequence changes alter the stability landscape. A simulation campaign optimized for target transfer should therefore choose descriptors whose physical meaning is expected to transfer to the experimental endpoint, rather than maximizing

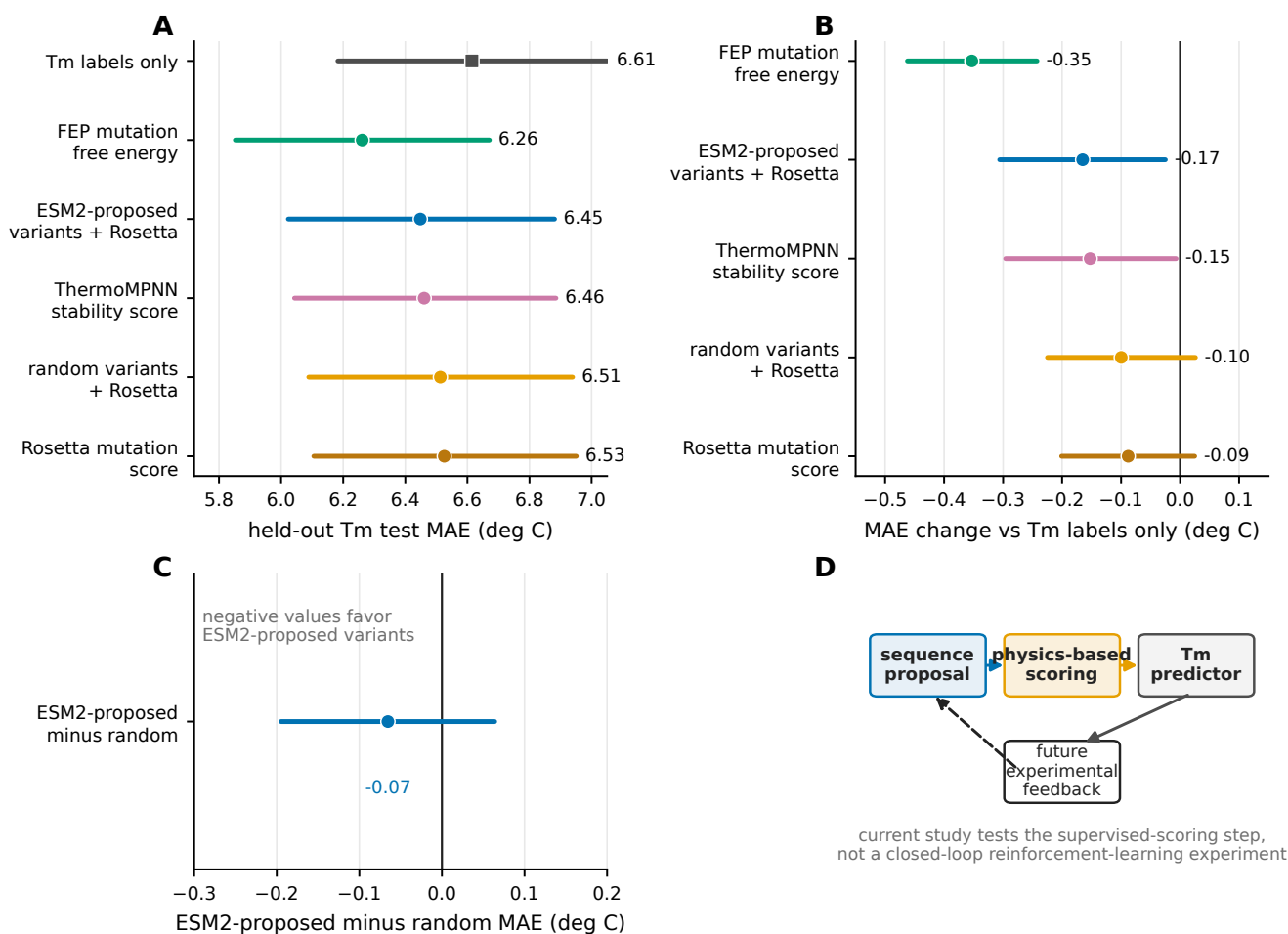


Fig. 4. Structure-based labels and language-model-proposed variants. (a) Final test MAE for Tm-only learning, FEP, and structure-based source labels. (b) Paired Δ MAE estimates relative to Tm-only learning. Negative values indicate improvement. (c) Direct paired comparison between ESM2-proposed variants scored by Rosetta and random variants scored by Rosetta. Negative values favor the ESM2-proposed set. (d) The proposal-set analysis motivates future tests of a generator–physics–predictor loop, but the present study evaluates only the supervised-scoring step and does not perform closed-loop optimization.

label count alone.

Physics-scored sequence proposals as a future direction

The Rosetta-scored proposal-set result is best viewed as a feasibility test for future design workflows. Rosetta scores assigned to ESM2-proposed variants improved Tm prediction relative to Tm-only learning, but the direct comparison with random variants remained exploratory because its confidence interval crossed zero. We therefore do not interpret this result as evidence that the current proposal model has already optimized nanobody thermostability. The useful point is more limited: sequences proposed by a protein language model can be annotated by a physics-based score and then used as supervised source data for a Tm predictor.

That supervised-scoring step is the only part of a design loop tested here. In a future closed loop, a sequence generator could propose variants, physics-based calculations could annotate them, the Tm predictor could update its estimate of experimental stability, and the resulting score could be used as a reward or selection signal for the

next round of proposals. Such a workflow would connect the present transfer-learning result to active learning or reinforcement learning, but those closed-loop steps and the experimental validation of newly proposed high-Tm nanobodies remain outside the scope of this paper.

Limitations

Several limitations should guide interpretation. First, the experimental Tm training set is small, and the conclusions should be tested on additional nanobody collections and other protein families. Second, the source-label comparison is limited to the computational labels available here. Other free-energy protocols, stability predictors, trajectory summaries, or structure-based descriptors may transfer differently. Third, the FEP curve does not establish a clean power-law scaling relation; it supports a beneficial trend at the largest tested setting, but not a universal scaling exponent. Fourth, no new experimental measurements were performed to validate high-stability candidates. Finally, raw simulation trajectories may be too large to deposit fully, so repro-

ducibility will depend on processed data, scripts, and clear reporting of the simulation protocol.

Outlook

For low-data experimental protein-property prediction, simulation labels should be treated as design choices rather than as generic data augmentation. The relevant question is not whether simulations can produce many labels, but whether those labels encode information that improves the target task under a target-centered validation protocol. In this study, mutation free-energy labels provided that information for nanobody T_m prediction. The broader strategy is therefore to use experimental data to define the target phenotype, use simulation to annotate physically meaningful perturbations around that phenotype, and evaluate the combined model only on held-out experimental data. This provides a practical path from simulation-assisted prediction toward future closed-loop protein engineering.

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AUTHOR CONTRIBUTIONS

Taihei Murakami and Kentaro Sasaki contributed equally to this work. Yasuhiro Matsunaga conceived and supervised the study, acquired funding, developed the analysis and training code, performed the analyses, interpreted the results, prepared the figures, and wrote the manuscript. Kentaro Sasaki performed computational calculations. Taihei Murakami contributed to draft writing. All authors reviewed and approved the final manuscript.

Declaration of competing interest

Taihei Murakami is an employee of Epsilon Molecular Engineering, Inc. Yasuhiro Matsunaga is a scientific advisor of Epsilon Molecular Engineering, Inc.

Data availability

Processed data required to reproduce the main and supplementary figures will be deposited in Zenodo at [ZENODO_DATA_DOI_PLACEHOLDER](#). Raw simulation trajectories are large and will be made available at [RAW_DATA_LOCATION_OR_REQUEST_POLICY_PLACEHOLDER](#).

Code availability

Analysis, training, and figure-generation code will be available on GitHub at [GITHUB_REPOSITORY_URL_PLACEHOLDER](#). An archival release will be deposited in Zenodo at [ZENODO_CODE_DOI_PLACEHOLDER](#).

AI-use disclosure

AI-assisted tools were used to support language editing, code development, and figure preparation. The authors

reviewed and take responsibility for all manuscript content, analyses, and conclusions.

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Supplementary Methods

Experimental target data and target-label scaling

All reported target-task comparisons use the same NbBench thermo-seq endpoint (15). The processed experimental tables contain an amino-acid sequence field and a melting-temperature label in degrees Celsius. The processed split sizes are 57 training sequences, 114 validation sequences, and 396 held-out test sequences.

Split definitions. The split definitions are fixed throughout the study. The 57 training examples are used to fit neural-network parameters. The validation split is used for checkpoint selection, early stopping, and selection among candidate model settings. The test split is held out until those choices have been fixed. The same rule is applied to Tm-only models and to models trained with computational source labels.

Target-label scaling. The training workflow transforms the experimental Tm values to a scaled regression target before optimization. The scaler is fit only on the 57 training labels, using a robust scaling step followed by min-max scaling to [0,1]. Validation labels are transformed with this training-fitted scaler during model selection. Test predictions are averaged across seeds in scaled-label space and are then inverse-transformed to degrees Celsius before error metrics are computed. The validation and test label distributions therefore do not influence the fitted target-label transform.

Experimental-label scaling reference. The experimental-label scaling reference uses the same training split but subsamples it to the requested number of labels with the run seed. The validation and test splits are never subsampled. This experiment is used only as a positive control for the analysis pipeline: if more true target labels are added, the Tm prediction error should improve.

Processed computational source-label tables

Computational labels are loaded as source tasks rather than appended to the Tm data set as additional Tm measurements. Table 1 summarizes the processed target and source files used in the reported comparisons. Mutation-effect source tables contain sequence fields and [0,1]-scaled source-label fields. The MD-derived Q-value table is stored in the same processed-table format so that it can enter the same source-task loader; for that source, the scaled label is a min-max scaled Q-value, not a mutation-free-energy label.

Table 1. Processed target and source-label data. Row counts exclude header rows.

Data type	Role	Processed rows	Use in reported comparisons
Experimental Tm	Target train	57	Model fitting
Experimental Tm	Target validation	114	Checkpoint and setting selection
Experimental Tm	Target test	396	Final held-out evaluation
FEP mutation free energy	Source tasks	435 + 409	Main source task
Rosetta mutation score	Source tasks	435 + 409	Source comparison
ThermoMPNN stability score	Source tasks	435 + 409	Source comparison
Random variants scored by Rosetta	Source tasks	1000 + 1000	Source comparison
ESM2-proposed variants scored by Rosetta	Source tasks	1000 + 1000	Source comparison
MD-derived Q-value	Source task	1143	Source comparison and scaling control

Source-task assignment. The data loader assigns each experimental or computational row to a target or source task. Experimental Tm rows use the Tm head. The two template-specific mutation-effect tables use separate source heads in the main mutation-effect model, and the MD-derived Q-value table uses a single source head. This assignment determines which scalar regression head is used and which task-specific loss term an example contributes to.

Sequence-overlap check. Exact sequence overlap was checked between the processed source-label tables and the fixed NbBench splits. The processed FEP/Rosetta mutation-effect tables, random variants scored by Rosetta, and ESM2-proposed variants scored by Rosetta had no exact sequence matches to the NbBench training, validation, or test sequences. The ThermoMPNN tables used the same template-specific variant design as the mutation-effect sources. The SAbDab-derived MD Q-value table contained exact matches to 2 NbBench training sequences, 4 validation sequences, and 8 test sequences. This overlap is disclosed because it affects interpretation of the MD control; it cannot explain the positive FEP result, and the MD Q-value source did not improve held-out Tm prediction.

Source-label sampling. For mutation-effect experiments with requested source-label count n , the loader samples up to n rows independently from each of the two template tables using the run seed. Thus, the final FEP setting with $n = 320$ supplies 320 sampled 1MEL rows and 320 sampled 4IDL rows before the source-task train/validation split. For MD-derived Q-value experiments, the loader samples n rows from the single processed Q-value table. Source rows are split 80/20 into source-training and source-validation rows using the same seed. The production training

call then filters the validation data passed to the trainer so that checkpoint selection and early stopping are based only on the experimental Tm validation examples.

SAbDab-derived nanobody structure panel for MD

The MD source-label panel was constructed from SAbDab, the Structural Antibody Database (17). The archived SAbDab nanobody summary table contains 2422 rows with PDB identifier, heavy-chain assignment, light-chain assignment, model index, antigen annotation, experimental method, resolution, species, and other SAbDab fields. The manifest construction workflow matches this summary against the archived IMGT-renumbered PDB files.

Structure-selection rule. The selection rule is deterministic. For each PDB identifier with both a SAbDab summary row and an IMGT-renumbered PDB file, the workflow picks the first heavy-chain row encountered in the SAbDab table. The chain is accepted only if the same chain identifier is present in the IMGT PDB file. The output manifest contains one entry per PDB identifier, with fields for the manifest row number, PDB identifier, selected heavy chain, SAbDab TSV row, IMGT PDB path, and MD run directory. This initial one-per-PDB manifest contained 1177 structures.

400 K simulation panel. The 400 K MD campaign reused the systems prepared in the 300 K campaign and created a 1146-entry 400 K manifest. The selected 400 K manifest consisted of 798 X-ray structures, 345 electron-microscopy structures, and 3 NMR structures according to the SAbDab method field. All selected rows lacked an annotated light chain in the SAbDab summary, consistent with a single-domain antibody panel. After trajectory and Q-value processing, 1143 unique PDB identifiers yielded usable source-label rows.

MD system preparation and simulation protocol

System preparation. Each selected structure was prepared as a nanobody-only MD system. The preparation workflow uses the IMGT-renumbered PDB as the structure source, extracts the selected heavy chain, includes only protein atoms, and leaves termini uncapped. The preparation call uses pH 7.4. Protein chains are cleaned with PDBFixer, and pH-aware protonation is assigned through pdb2pqr/PROPKA when available (18, 19); its fallback path uses AmberTools-based hydrogen assignment and Amber naming. The resulting prepared chain is solvated with explicit OPC water, 15 Å padding, and 0.15 M NaCl. Amber topology and coordinate files are built with ff19SB and OPC (20–22).

300 K equilibration branch. The base 300 K simulation branch uses OpenMM on CUDA devices (23). It runs 1 ns of NVT equilibration followed by 1 ns of NPT equilibration at 300 K and 1 atm, with a 2-fs timestep and no hydrogen-mass repartitioning. Production at 300 K was generated for the broader MD campaign, but the source label used in this manuscript comes from the 400 K branch.

400 K production branch. The first 400 K relaxation stage restarts from the 300 K equilibrated state and runs 1 ns of NVT at 400 K with C α restraints. The second relaxation node restarts from that state and repeats 1 ns of 400 K NVT with C α restraints. Both relaxation stages use a 2-fs timestep and no hydrogen-mass repartitioning. The production node runs 100 ns of restraint-free 400 K NVT, with hydrogen-mass repartitioning, 4 amu hydrogen mass, a 4-fs timestep, and coordinate/energy output every 100 ps.

Integrator and reporting settings. In the OpenMM implementation, explicit-solvent periodic systems use PME electrostatics with a 1.0-nm nonbonded cutoff and constraints on bonds involving hydrogen atoms. NPT stages use a Monte Carlo barostat. NVT stages omit the barostat. The integrator is the Langevin middle integrator with friction 1 ps⁻¹. The 100-ps reporting interval gives 1000 expected frames for a complete 100 ns production trajectory.

MD-derived Q-value extraction

Trajectory input requirements. The processed MD source table was generated from the archived 400 K production trajectories. A system was eligible only if it contained a production trajectory, a prepared reference structure, and an Amber parameter file. The amino-acid sequence was extracted from the prepared structure by reading the first C α atom for each residue. Sequences shorter than 50 residues were excluded.

Native-contact definition. Q-values are computed with MDTraj (24). The topology is loaded from the Amber parameter file, and the native reference is frame 0 of the production trajectory. The contact calculation uses protein backbone heavy atoms. Candidate native contacts must be separated by more than three residues, have a reference-frame distance below 0.45 nm, and include at least one atom from a hydrophilic residue (Asp, Glu, Gln, Asn, Arg, Lys, or His, including Amber protonation variants). For a native-contact pair (i, j) with reference distance r_{ij}^0 and instantaneous distance r_{ij} , the per-frame contribution is

$$q_{ij} = \frac{1}{1 + \exp\{\beta(r_{ij} - \lambda r_{ij}^0)\}},$$

with $\beta = 50 \text{ nm}^{-1}$ and $\lambda = 1.8$, following the smooth native-contact form used in Best–Hummer-style Q-value analyses (25). The frame-level Q-value is the mean of q_{ij} over all selected native contacts.

Trajectory-level Q-value label. The trajectory-level label is the mean Q-value over the terminal portion of the production trajectory. For trajectories with more than 300 frames, the implementation uses the larger of 300 frames and one third of the available frames; for shorter trajectories it uses all available frames. For complete 100 ns trajectories with 100-ps output, this corresponds to 333 terminal frames, approximately the final 33 ns. In the processed 400 K Q-value table, the number of frames used ranges from 66 to 333 with median 333. The number of native contacts ranges from 28 to 728 with median 173. The sequence length range is 58–461 residues with median 122. Raw Q-values range from 0.0437 to 0.9994. These raw Q-values are min–max scaled to $[0,1]$ for source-task training.

Sequence tokenization and encoder inputs

Sequences are tokenized with the ESM2 tokenizer associated with the selected encoder (8). The workflow uses truncation and fixed-length padding to 160 residues. For the production protocol, the trainer receives all target and sampled source rows for training, but its evaluation data are filtered to the experimental Tm validation rows.

Hot-encoder input path. The hot-encoder setting keeps tokenized sequence inputs in the training data and updates the ESM2 weights. In the frozen-encoder setting, ESM2 is run once for each target and source sequence to precompute first-token embeddings, and the subsequent optimizer updates only the regression trunk and task heads.

Neural-network architecture

The default encoder is the 8M-parameter ESM2 model. The model takes the first-token ESM2 representation as the sequence-level embedding. The shared regression trunk has dimensions hidden-size $\rightarrow 256 \rightarrow 128 \rightarrow 32$, with ReLU activations and dropout after the first two hidden layers. The Tm head maps the 32-dimensional representation to one scaled scalar prediction.

Target and source heads. The Tm-only baseline consists of the ESM2 encoder, the shared trunk, and the Tm head. Source-transfer models keep this target path and add source heads. The main mutation-effect model uses separate source heads for the two template-specific tasks. The code also implements source-head controls with a single shared mutation-effect head, a template-conditioned head, and a template-specific affine calibration on top of a shared head. These controls test whether the two template-specific mutation-effect sources require separate output calibration.

Architecture sensitivity analyses. Architecture controls include residual, latent, and mixture-style source-fusion variants. In these variants, the source path can produce a correction to the Tm prediction, a latent interaction between Tm and source features, or a gate between a target-only expert and a source-informed expert. For the main reported FEP result, the selected model is the original shared-trunk architecture with separate template-specific source heads. The architecture controls are retained as sensitivity analyses rather than as the main claim.

Training objective, optimizer, and checkpoint selection

Training examples are represented by sequence, scaled label, and task identifier. Each example contributes loss only through the regression head corresponding to its task. The per-task loss is Huber loss with $\delta = 1.0$ on the scaled label. The default multi-task objective uses learned task-uncertainty weights (38). For task k with loss L_k and learned log-scale s_k , the loss contribution is

$$\frac{L_k}{2\exp(2s_k)} + s_k.$$

The total loss is the sum of the contributions from the tasks present in the minibatch. Fixed source-loss weights are also implemented and were included in the candidate-setting searches for source classes where loss balance could change validation performance.

Optimizer and schedule. The optimizer is AdamW. The default head/trunk learning rate is 3×10^{-4} , the default hot-encoder learning rate is 1×10^{-4} , weight decay is 0.04, dropout is 0.15, batch size is 16, warmup is 100 steps, and the schedule is cosine decay. Training runs for at most 400 epochs. Checkpoints are evaluated and saved once per epoch. Mixed precision is enabled when CUDA is available. In the hot-encoder setting, the custom trainer creates separate optimizer parameter groups for ESM2 parameters and newly initialized trunk/head parameters. In the frozen-encoder setting, ESM2 parameters are non-trainable and no encoder optimizer group is used.

Checkpoint selection. Within each run, the best checkpoint is the checkpoint with the lowest mean squared error on the experimental Tm validation rows. Early stopping monitors the same validation metric with patience 15 and threshold 0. Source-task validation rows are not passed to the trainer in the production protocol. This implementation detail is central for the comparison: source labels influence the learned representation through training loss, but they do not decide which checkpoint is used for target-task prediction.

Candidate-setting selection and final evaluation

Candidate settings are selected on the experimental Tm validation split. The common candidate grid includes the default setting; encoder learning rates of 3×10^{-5} , 1×10^{-4} , and 3×10^{-4} ; a lower head learning rate of 1×10^{-4} paired with encoder learning rate 3×10^{-5} ; and dropout rates 0.05, 0.15, and 0.30. Fixed source-loss weights are added for source classes where the source label scale may require explicit balancing. Tm-only candidates include shared, residual, and latent architecture variants. Mutation-effect source candidates use the shared-trunk family for the main source comparison, with a separate source-head design comparison. MD-derived source candidates include fixed-weight settings because Q-value labels differ in scale and interpretation from mutation-free-energy labels.

Seeded candidate evaluation. Each candidate is first evaluated as a three-seed ensemble on the Tm validation split. The selected setting for each source class is then evaluated on the held-out test split as a ten-seed ensemble. Final comparisons use seeds 1–10. The run seed controls Python, NumPy, PyTorch, and CUDA randomness when CUDA is available; it also controls target-training subsampling in the target-label scaling experiment, source-row sampling, and source train/validation splitting. The label-count curves use fixed source-specific training recipes while the number of source labels is varied.

Statistics and stored outputs

The primary metric is MAE in degrees Celsius on the 396 held-out test examples. For each condition, the ten seeded models produce scaled Tm predictions for every test sequence. Predictions are averaged in scaled-label space and then inverse-transformed to degrees Celsius. Root mean squared error, coefficient of determination, Spearman correlation, and Pearson correlation are also computed by the evaluation code, but the figures emphasize MAE because it is directly interpretable as average temperature error.

Bootstrap intervals. Single-condition confidence intervals are estimated by nonparametric bootstrap resampling over test examples. The absolute-error vector from the ten-seed ensemble is resampled with replacement 10,000 times using a fixed bootstrap seed. The reported 90% interval is the 5th to 95th percentile range of the bootstrapped MAE values. Paired comparisons use the same bootstrap indices for the candidate and reference absolute-error vectors. The reported Δ MAE is candidate minus reference, so negative values indicate lower error than the reference.

Stored run summaries. Each training and evaluation call writes a structured run summary that records command-line arguments, relevant environment variables, resolved model settings, per-label-count metrics, bootstrap intervals, per-example absolute errors, and paired bootstrap summaries. Compact summary tables derived from these run summaries provide the inputs for the main source comparison, frozen-encoder controls, source-head controls, and source-combination controls.

Interpreting the label-count curves

Target-label scaling reference. The target-label scaling curve is a positive-control check on the pipeline. It asks whether the same model family and evaluation code detect the expected gain when more true experimental Tm labels are available. The FEP source-label curve asks a different question: whether more labels from a physically related but different computed quantity can improve the same held-out Tm endpoint. The MD Q-value curve asks whether adding more simulation-derived labels is sufficient when the source label measures a more indirect structural response. The important comparison is therefore not simply “simulation versus no simulation.” It is whether the source label carries the kind of physical information that transfers to the target property under target-centered model selection.

Supplementary figure source tables

The supplementary figures are generated from tabular source files derived from the same run summaries used for the main figures. The processed source tables, rendered supplementary figures, and figure-generation code will be distributed with the public analysis archive.

Fig.-to-data and code mapping

The current manuscript figures are regenerated from compact result summaries and source tables rather than from scratch run directories. The mapping below is the first place to check when auditing or extending a figure.

Generated supplementary tables. The final source-screen settings used in Figs. 3 and 4 and the frozen-encoder controls are recorded in generated supplementary tables. These tables include the selected architecture, validation-selected setting, source-label count, validation MAE, held-out test MAE, bootstrap interval, and the run-summary reference for each condition. A panel-level mapping table for all supplementary figures is included in the public analysis archive.

Table 2. Manuscript Fig.-to-data mapping. The table identifies the processed summaries and scripts used to regenerate each display item.

Display item	Inputs	Script	Primary comparison
Fig. 1	Conceptual protocol	Main-figure script	Target-centered transfer-learning design
Fig. 2	Main scaling summaries	Main-figure script	Label-count scaling for Tm labels, FEP labels, and MD Q-value labels
Fig. 3	Final hot- and frozen-encoder source-screen summaries	Main-figure script	All-atom FEP and MD labels versus training only on experimental Tm labels
Fig. 4	Final source-screen summary	Main-figure script	Rosetta, ThermoMPNN, and Rosetta-scored proposal sets
Supplementary Figs. 1–5	Supplementary manifest and generated TSV tables	Supplementary-figure script	Data-set composition, candidate-setting searches, model controls, scaling controls, and MD-feature controls

Supplementary Figs.

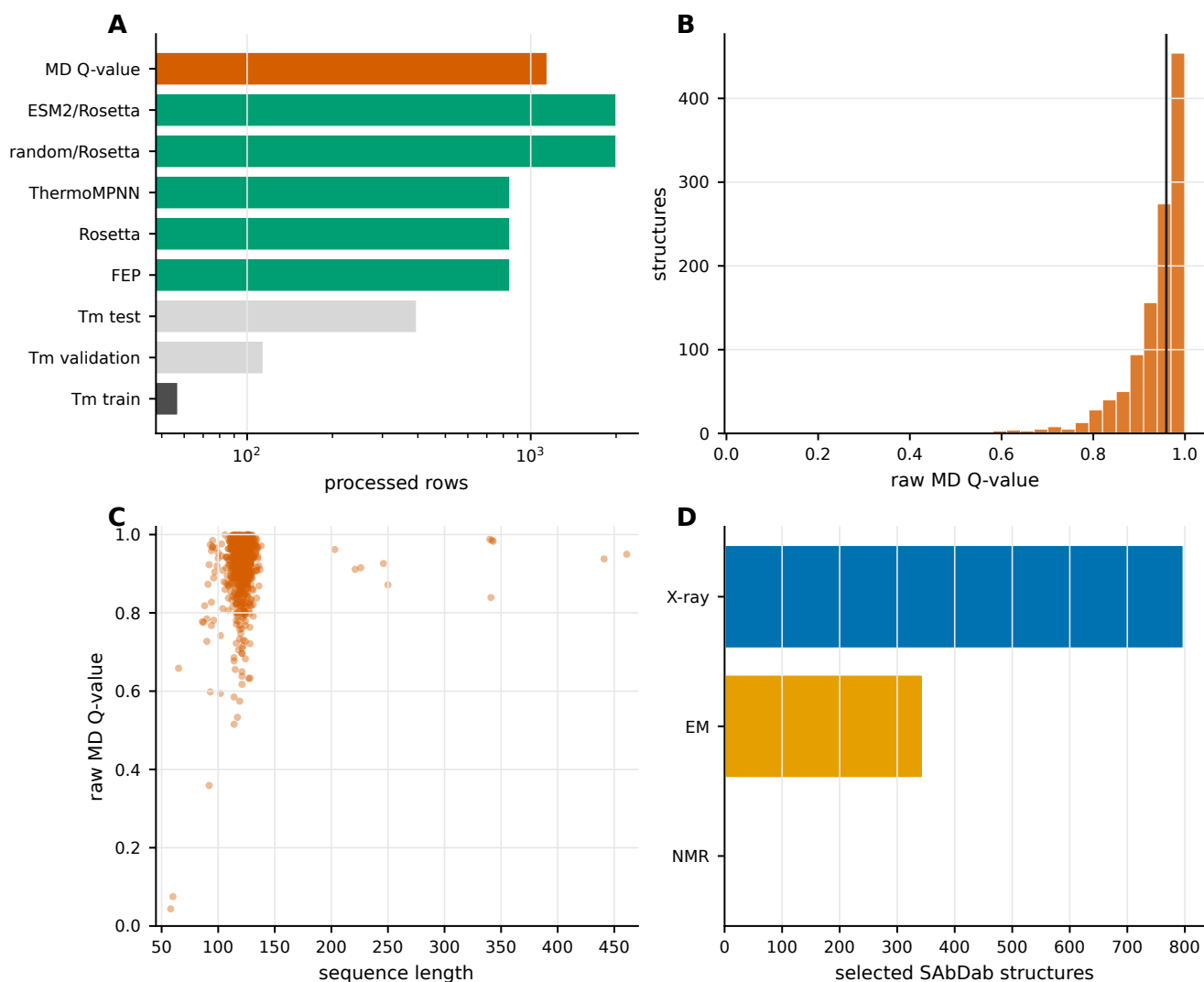


Fig. 5. Supplementary Fig. 1. Processed data sets and MD-derived Q-value labels. (a) Processed target and source-label row counts used in the reported experiments. (b) Distribution of raw MD-derived Q-values before min-max scaling. (c) Relationship between sequence length and raw MD-derived Q-value. (d) Experimental methods for the selected SABDab structures used in the 400 K MD panel.

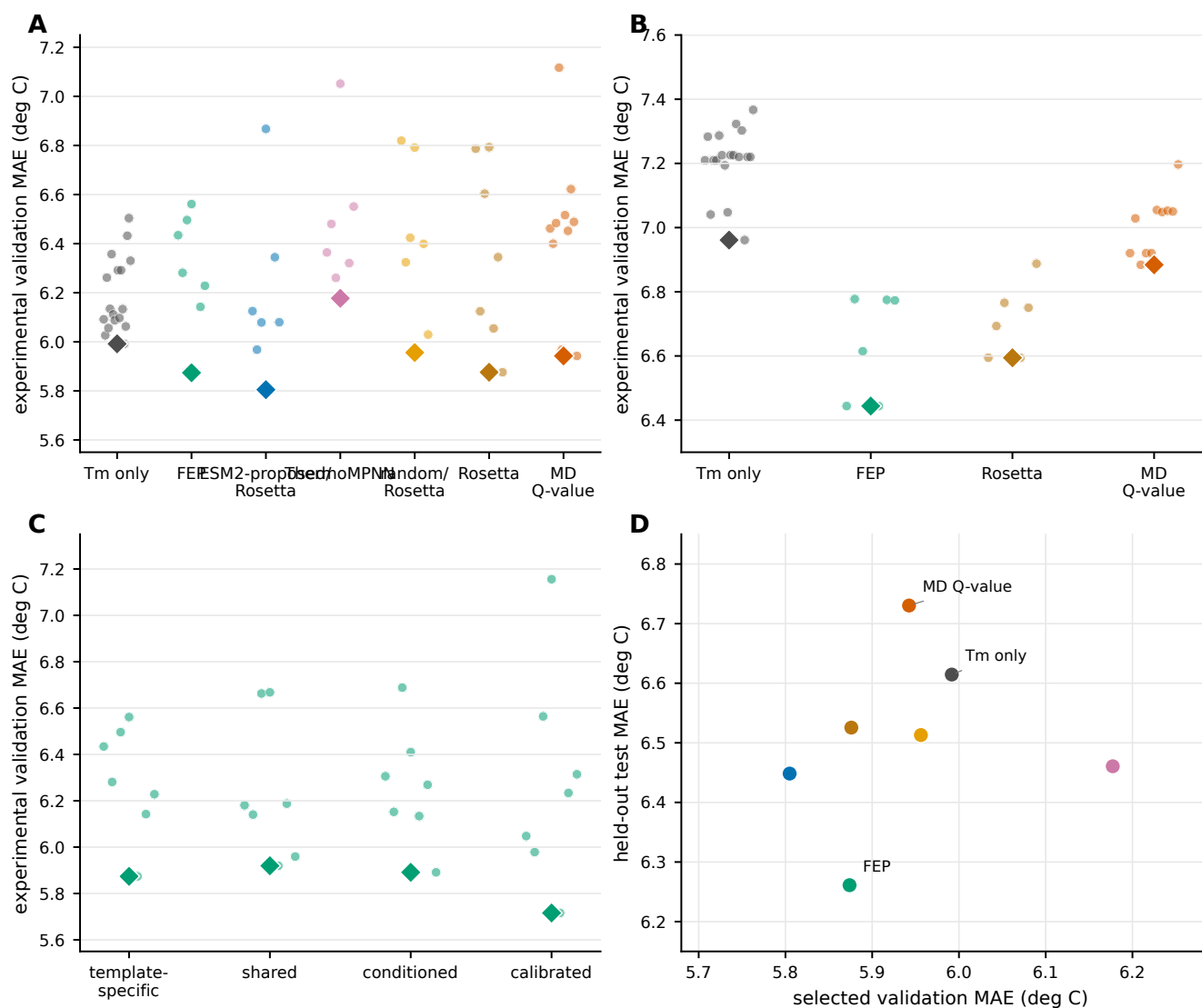


Fig. 6. Supplementary Fig. 2. Candidate-setting searches and target-validation selection. (a) Experimental validation MAE values for hot-encoder source comparisons. (b) Experimental validation MAE values for frozen-encoder controls. (c) Candidate source-head designs for FEP mutation free-energy labels. (d) Relationship between selected validation MAE and held-out test MAE for the final source comparison. Diamonds mark the best validation setting in panels a–c.

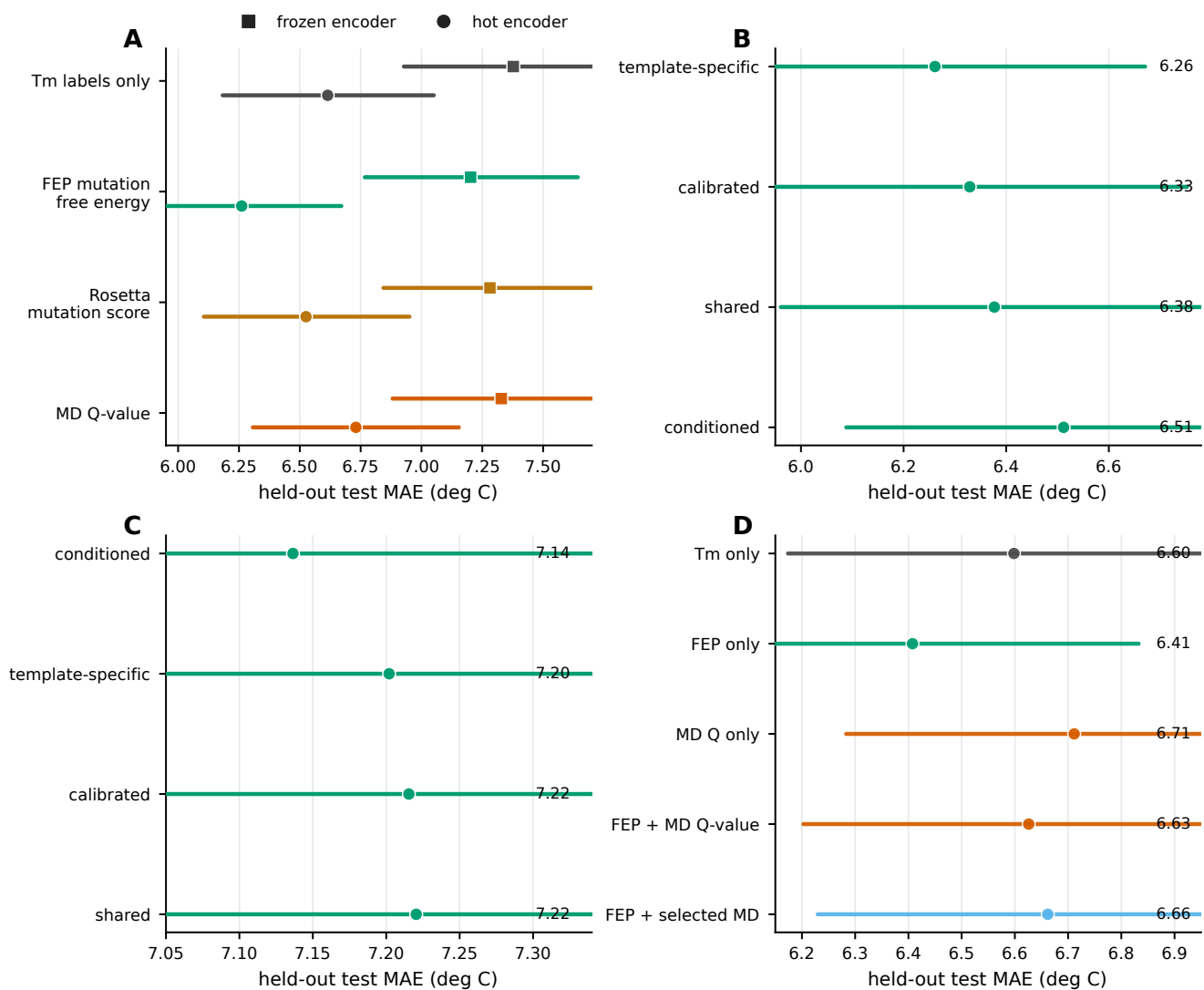


Fig. 7. Supplementary Fig. 3. Model and source-combination controls. (a) Frozen-encoder and hot-encoder comparisons for selected source labels. (b) Hot-encoder FEP source-head controls. (c) Frozen-encoder FEP source-head controls. (d) Source-combination controls comparing Tm-only, FEP-only, MD Q-value-only, and combined-source settings.

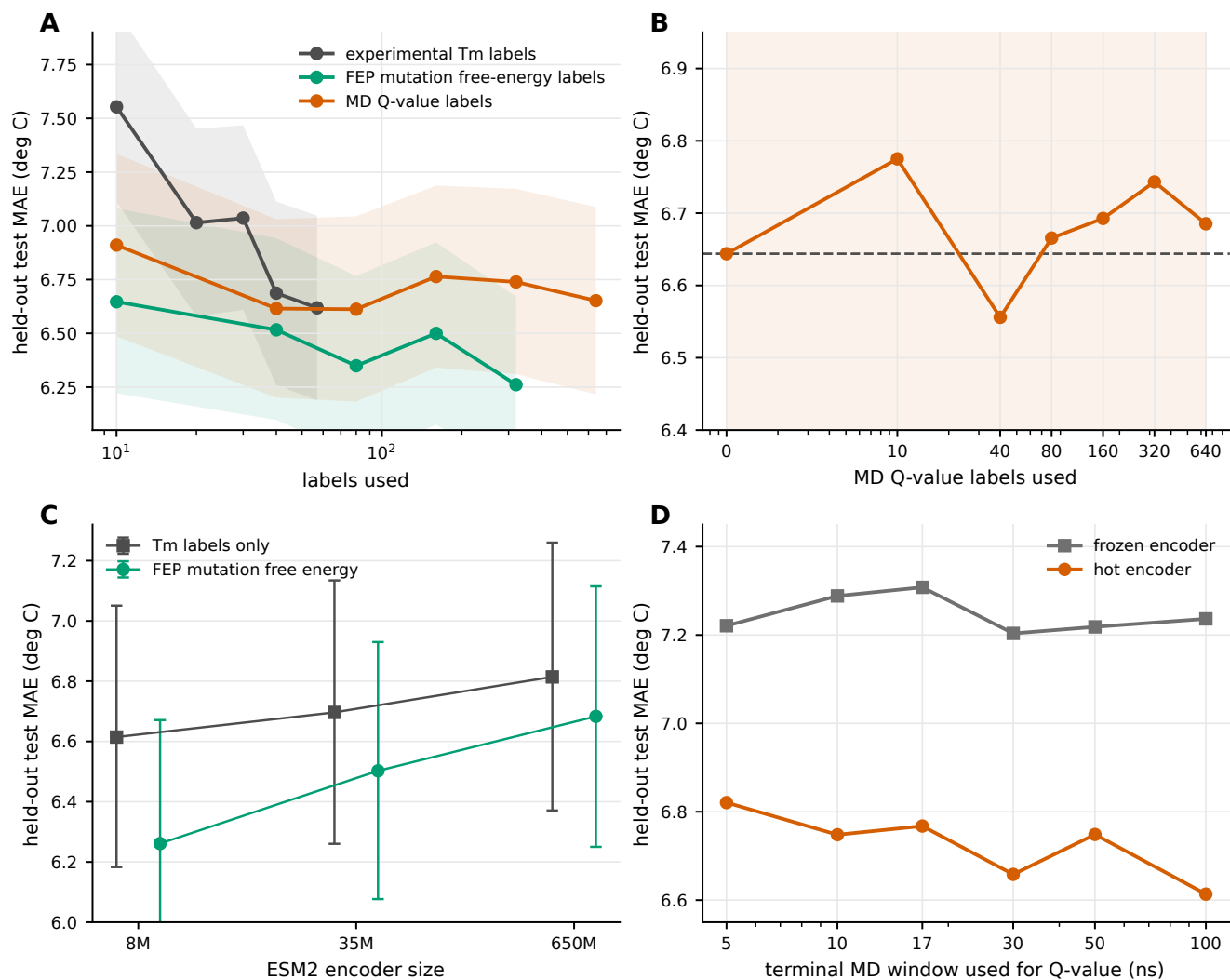


Fig. 8. Supplementary Fig. 4. Scaling, encoder-size, and MD-window controls. (a) Label-count curves for experimental Tm labels, FEP labels, and MD Q-value labels. (b) MD Q-value label-count curve after selecting the candidate setting separately for each label count on the experimental validation split. (c) ESM2 encoder-size controls for Tm-only and FEP settings. (d) MD trajectory-window controls for Q-value labels.

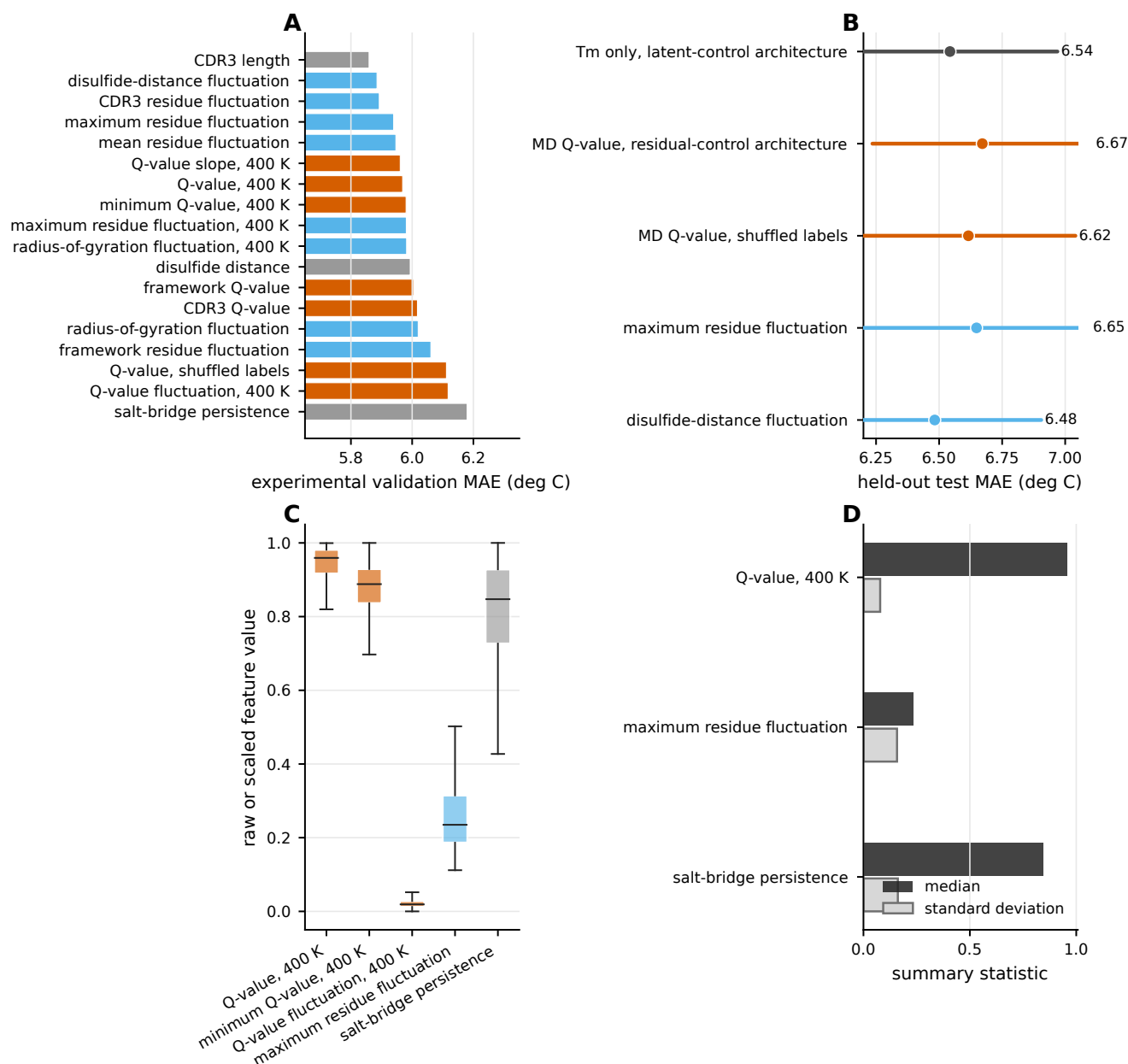


Fig. 9. Supplementary Fig. 5. MD feature and architecture controls. (a) Experimental validation MAE values for alternative MD-derived features. (b) Held-out test MAE values for selected architecture and shuffled label controls. (c) Distributions of representative MD-derived feature values. (d) Summary statistics for representative MD-derived feature values.